| | |
|---|---|
| United States Patent | 12409543 |
| Kind Code | B1 |
| Date of Patent | September 09, 2025 |
| Inventor(s) | Rivera; Benjamin C. |


## Lock for folding tool


### Abstract

A lock is provided for a folding tool including a lock that engages a folding tool in an unfolded, deployed position to prevent unintended folding of the tool. By way of example a tool is provided including: a handle portion; a folding tool portion, where the folding tool portion is configured to be rotatable between a folded, stowed position at least partially within the handle portion, and a deployed position substantially outside of the handle portion; and a locking element, where the locking element is configured to engage a locking surface of the folding tool portion in response to the folding tool portion moving to the deployed position, where the locking element includes an arcuate surface for engaging the locking surface of the folding tool portion.



| | |
|---|---|
| **Inventors:** | **Rivera; Benjamin C. (Portland, OR)** |
| **Applicant:** | **Leatherman Tool Group, Inc.** (Portland, OR) |
| **Family ID:** | **95942335** |
| **Assignee:** | **LEATHERMAN TOOL GROUP, INC. (Portland, OR)** |
| **Appl. No.:** | **18/813832** |
| **Filed:** | **August 23, 2024** |

### Publication Classification

**Int. Cl.:** **B25G3/38** (20060101); **B25F1/04** (20060101); **B26B1/04** (20060101)

**U.S. Cl.:**

CPC **B25G3/38** (20130101); B25F1/04 (20130101); B26B1/042 (20130101)

### Field of Classification Search

**CPC:** B25G (3/38); B25F (1/04); B26B (11/042); B26B (11/044)

**USPC:** 81/490

## References Cited

**U.S. PATENT DOCUMENTS**

| Patent No. | Issued Date | Patentee Name | U.S. Cl. | CPC |
|---|---|---|---|---|
| 4404748 | 12/1982 | Wiethoff | 30/161 | B26B 1/046 |
| 5822866 | 12/1997 | Pardue | 30/160 | B26B 1/048 |
| 7066060 | 12/2005 | Sebileau | 81/3.43 | B67B 7/184 |
| 7854067 | 12/2009 | Lake | 30/159 | B26B 1/046 |

*Primary Examiner:* Shakeri; Hadi

*Attorney, Agent or Firm:* ALSTON & BIRD LLP

## Background/Summary

TECHNOLOGICAL FIELD

(1) An example embodiment relates generally to a lock for a folding tool, and in one embodiment, a lock that engages a folding tool in an unfolded, deployed position to prevent unintentional folding of the tool.

BACKGROUND

(2) Folding tools, such as pocketknives, multipurpose tools, etc. are widely popular for their utility in a number of different applications. Pocketknives generally have a folding blade that is moved from a stowed position within a handle of the pocketknife to a deployed position where the folding blade is extended and can be used. A multipurpose tool includes a number of tool members carried by a common frame. A multipurpose tool may include different combinations of tool members depending upon its intended application. For example, multipurpose tools that are designed for a more universal or generic application can include pliers, a wire cutter, a bit driver, one or more knife blades, a saw blade or the like.

(3) One reason for the popularity of folding tools is the convenience of a tool that can be carried in a small form factor, such as in a pocket. Further, the ability to fold a tool within a handle generally provides a small package that does not have sharp edges that may be present on the tool folded within the handle. For pocketknives, the folded, stowed position provides a safe way to carry a dangerous sharp object without requiring a sheath or the length of the entire handle and blade to be carried. Multipurpose tools provide a wide range of functionality with a single tool, thereby reducing the need to carry a number of different tools to perform the same functions. For example, a single multipurpose tool may be carried instead of a pair of pliers, one or more screwdrivers, a knife and a bottle opener. These different tools can be folded within the handle or handles of the multitool and have similar benefits as a pocketknife embodiment. As such, the burden placed upon the user is reduced since the user need only carry a single multipurpose tool.

(4) As pocketknives and multipurpose tools are frequently carried by users in the field, it is desirable for the pocketknives and multipurpose tools to be relatively small and lightweight, while remaining rugged so as resist damage. In order to reduce the overall size of a tool, some tools have been designed to be foldable.

BRIEF SUMMARY

(5) A tool, such as folding tool is provided which may include, for example, a pocketknife, a

multipurpose tool, etc. Embodiments generally relate to a lock for a folding tool, and in one embodiment, a lock that engages a folding tool in an unfolded, deployed position to prevent unintended folding of the tool. Embodiments described herein include tool including: a handle portion; a folding tool portion, wherein the folding tool portion is configured to be rotatable about a folding tool axis between a folded, stowed position at least partially within the handle portion, and a deployed position substantially outside of the handle portion, where the folding tool portion comprises a cam portion disposed about the folding tool axis, where the cam portion defines a locking surface; and a locking element, where the locking element is configured to engage the locking surface of the folding tool portion in response to the folding tool portion moving to the deployed position, where the locking element is spring biased by a spring in a direction parallel to the folding tool axis, and where the locking element defines at least one of an arcuate surface or a wedge surface for engaging the locking surface of the folding tool portion.

(6) According to some embodiments, the tool further includes a button, where the button is at least one of attached to the locking element or integrally formed with the locking element, where the button is configured to drive the locking element against spring bias to disengage the locking element from the locking surface. The button of an example embodiment defines a central button axis extending therethrough parallel to the folding tool axis, where the locking element defines a central locking element axis parallel to the folding tool axis, and where the central button axis and the central locking element axis are offset from one another.

(7) According to certain embodiments the handle portion defines a first handle piece and a second handle piece, where the first handle piece defines a first aperture, where the second handle piece defines a second aperture, where the locking element is aligned with and guided by the first aperture, where the button is aligned with and guided by the second aperture. The button and the locking element of some embodiments are not mechanically coupled to the handle portion or the spring. According to some embodiments the locking element defines a detent ball, where the cam portion defines a recess, wherein in the folded, stowed position the detent ball engages the recess and holds the folding tool portion in the folded, stowed position.

(8) According to some embodiments, the handle portion includes an abutment and the folding tool portion comprises a stopping surface defined by the cam portion, where in the deployed position, the folding tool portion is secured rotationally about the folding tool axis in a first direction by the locking element engaging the locking surface and in a second direction by the stopping surface engaging the abutment. In the deployed position of some embodiments, the locking element is held between the locking surface of the folding tool portion and a stop surface of the abutment. According to some embodiments the handle portion includes a first handle piece and a second handle piece, where the abutment is integrally formed with the first handle piece. According to certain embodiments the locking element defines the arcuate surface, wherein the arcuate surface of the locking element engages the locking surface of the folding tool portion at different angles based on a degree of overlap between the locking surface and the arcuate surface.

(9) Embodiments provided herein include a tool including: a handle portion including a first handle piece and a second handle piece, the first handle piece defining an abutment; a folding tool portion, where the folding tool portion is configured to be rotatable about a folding tool axis between a folded, stowed position at least partially between the first handle piece and the second handle piece, and a deployed position substantially outside of the handle portion, where the folding tool portion includes a cam portion disposed about the folding tool axis, where the cam portion defines a locking surface; and a locking element, where the locking element is configured to be captured between the locking surface of the folding tool portion and the abutment in response to the folding tool portion moving to the deployed position, where the locking element is spring biased by a spring in a direction parallel to the folding tool axis, where the locking element defines at least one of an arcuate surface or a wedge surface for engaging the locking surface of the folding tool portion.

(10) The tool of some embodiments includes a button, where the button is at least one of attached to the locking element or integrally formed with the locking element, where the button is configured to drive the locking element against spring bias to disengage the locking element from the locking surface. According to some embodiments the button defines a central button axis extending therethrough parallel to the folding tool axis, where the locking element defines a central locking element axis parallel to the folding tool axis, and where the central button axis and the central locking element axis are offset from one another.
(11) According to certain embodiments the handle portion defines a first handle piece and a second handle piece, where the first handle piece defines a first aperture, where the second handle piece defines a second aperture, where the locking element is aligned with and guided by the first aperture, and wherein the button is aligned with and guided by the second aperture. The button and the locking element of some embodiments are not mechanically coupled to the handle portion or the spring. According to some embodiments the locking element defines a detent ball, where the cam portion defines a recess, where in the folded, stowed position the detent ball engages the recess and holds the folding tool portion in the folded, stowed position.
(12) Embodiments provided herein include a tool including: a handle portion including a first handle piece defining a first aperture and a second handle piece defining a second aperture; a folding tool portion, where the folding tool portion is configured to be rotatable about a folding tool axis between a folded, stowed position at least partially between the first handle piece and the second handle piece, and a deployed position substantially outside of the handle portion, where the folding tool portion includes a cam portion disposed about the folding tool axis, where the cam portion defines a locking surface; a locking element, where the locking element is received within the first aperture where the locking element is movable along a locking axis parallel to the folding tool axis between an engaged position in which the locking element is engaged with the locking surface of the folding tool portion in the deployed position, and a disengaged position where the locking element does not contact the locking surface; and a button, where the button is received within the second aperture and is accessible through the second handle piece, where the button is at least one of attached to or integrally formed with the locking element, and where the button is movable along an axis parallel to the locking axis against a spring bias to move the locking element to the disengaged position.
(13) According to some embodiments the button defines a central button axis extending therethrough parallel to the folding tool axis, where the locking element defines a central locking element axis parallel to the folding tool axis, and where the central button axis and the central locking element axis are offset from one another. According to certain embodiments the locking element defines a detent ball, where the cam portion defines a recess, where in the folded, stowed position the detent ball engages the recess and holds the folding tool portion in the folded, stowed position. According to some embodiments a surface of the locking element that engages the locking surface comprises a surface texture configured to engage the locking surface with a higher friction than a smooth surface texture.
(14) Embodiments provided herein include a tool including: a handle portion; a folding tool portion, where the folding tool portion is configured to be rotatable between a folded, stowed position at least partially within the handle portion, and a deployed position substantially outside of the handle portion; and a locking element, where the locking element is configured to engage a locking surface of the folding tool portion in response to the folding tool portion moving to the deployed position, where the locking element includes an arcuate surface for engaging the locking surface of the folding tool portion.
(15) According to some embodiments the folding tool portion rotates between the folded, stowed position and the deployed position about an axis where the tool portion includes a cam portion disposed about the axis. According to certain embodiments, the cam portion defines the locking surface where the locking surface is substantially parallel to the axis. The locking element of some

embodiments is spring biased into engagement with the locking surface. The tool of some embodiments further includes a button, where the button is attached to the locking element and configured to drive the locking element against spring bias to disengage the locking element from the locking surface.

(16) According to some embodiments the handle portion includes an abutment and the folding tool portion includes a stopping surface defined by the cam portion, where in the deployed position, the folding tool portion is secured rotationally about the axis in a first direction by the locking element engaging the locking surface and in a second direction by the stopping surface engaging the abutment. According to certain embodiments the folding tool portion includes a stop surface defined by the cam portion where rotation of the folding tool portion to the folded, stowed position is stopped by the stop surface engaging with the abutment.

(17) The cam portion of certain embodiments defines a detent and the locking element includes a detent ball where in the folded, stowed position the detent ball engages the detent. According to certain embodiments the handle portion includes an abutment and the folding tool portion includes a stop surface defined by the cam portion where rotation of the folding tool portion to the folded, stowed position is stopped by the stop surface engaging with the abutment. The arcuate surface of the locking element of some embodiments engages the locking surface of the tool portion at different angles based on a degree of overlap between the locking surface and the arcuate surface.

(18) Embodiments provided herein include a tool including: a handle portion; a folding tool portion including a cam portion, where the cam portion is disposed about an axis of rotation of the folding tool portion, where the folding tool portion is configured to be rotatable about the axis between a folded, stowed position at least partially within the handle portion, and a deployed position substantially outside of the handle portion; and a locking element, where the locking element is configured to engage a locking surface of the cam portion in response to the folding tool portion moving to the deployed position, where the locking element defines an arcuate surface for engaging the locking surface of the folding tool portion, and where the locking element includes a retaining feature to retain the folding tool portion in the folded, stowed position.

(19) According to some embodiments the retaining feature includes at least one of a detent ball or a protrusion, where the cam portion defines a detent, and where in the folded, stowed position the at least one of the detent ball or the protrusion engages the detent. The cam portion of some embodiments defines the locking surface, where the locking surface is substantially parallel to the axis. According to certain embodiments the locking element is spring biased into engagement with the locking surface.

(20) The tool of some embodiments also includes a button, where the button is attached to the locking element and configured to drive the locking element against spring bias to disengage the locking element from the locking surface. According to some embodiments the handle portion includes an abutment and the folding tool portion includes a stopping surface defined by the cam portion, where in the deployed position, the folding tool portion is secured rotationally about the axis in a first direction by the locking element engaging the locking surface and in a second direction by the stopping surface engaging the abutment.

(21) According to certain embodiments the folding tool portion includes a stop surface defined by the cam portion, where rotation of the folding tool portion to the folded, stowed position is stopped by the stop surface engaging with the abutment. The handle portion of some embodiments includes an abutment, and the folding tool portion includes a stop surface defined by the cam portion, where rotation of the folding tool portion to the folded, stowed position is stopped by the stop surface engaging with the abutment. According to some embodiments the arcuate surface of the locking element engages the locking surface of the tool portion at different angles based on a degree of overlap between the locking surface and the arcuate surface. According to certain embodiments the folding tool portion further includes a grip, where the grip is actuatable by a thumb of a user's hand that holds the handle portion for one-handed deployment of the folding tool portion.

## Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Having thus described embodiments of the invention in general terms, reference will now be made to the accompanying drawings, which are not necessarily drawn to scale, and wherein:
(2) FIG. **1** illustrates a folding tool including a handle portion and a folding blade shown in the deployed position according to an example embodiment of the present disclosure;
(3) FIG. **2** illustrates the folding tool of FIG. **1** with the folding blade shown in the folded, stowed position within the handle portion according to an example embodiment of the present disclosure;
(4) FIG. **3** illustrates the tool of FIGS. **1** and **2** from a different view where a portion of the locking mechanism is visible within the handle portion according to an example embodiment of the present disclosure;
(5) FIG. **4** illustrates a folding tool including a handle portion and a folding blade shown in the deployed position with a piece of the handle portion removed according to an example embodiment of the present disclosure;
(6) FIG. **5** illustrates another view of the folding tool of FIG. **4** according to an example embodiment of the present disclosure;
(7) FIG. **6** illustrates another view of the tool of FIG. **4** in a section view, with the section line taken through a center of the button with the blade omitted according to an example embodiment of the present disclosure;
(8) FIG. **7** illustrates a detail view of the folding tool of FIG. **4** depicting the locking wedge in a locked position according to an example embodiment of the present disclosure;
(9) FIG. **8** illustrates a detail view of the folding tool of FIG. **4** depicting the locking wedge in an unlocked position and the spring omitted for ease of understanding according to an example embodiment of the present disclosure;
(10) FIG. **9** illustrates a detail view similar to that of FIG. **8** with the locking wedge in the locked, engaged position according to an example embodiment of the present disclosure;
(11) FIG. **10** illustrates a top view of a folding tool with the locking wedge in a locked, engaged position according to an example embodiment of the present disclosure;
(12) FIG. **11** illustrates another view of a locking tool including a locking wedge in a locked, engaged position with a locking surface of a folding blade according to an example embodiment of the present disclosure;
(13) FIG. **12** is a detail view of FIG. **11** from a different perspective depicting the engagement between the locking wedge and the locking surface according to an example embodiment of the present disclosure;
(14) FIG. **13** illustrates a detail view of the detail circle of FIG. **12** according to an example embodiment of the present disclosure;
(15) FIG. **14** illustrates a view of a folding tool including a folding blade and a handle portion with a different piece removed for understanding according to an example embodiment of the present disclosure;
(16) FIG. **15** illustrates a detail view of the components used to lock a folding blade securely in a deployed position according to an example embodiment of the present disclosure;
(17) FIG. **16** illustrates a view of a folding tool including a folding blade and a handle with a piece removed and the locking wedge in the unlocked position as the folding blade begins to rotate to the folded, stowed position according to an example embodiment of the present disclosure;
(18) FIG. **17** illustrates the view of FIG. **16** with the folding blade rotated further to the folded, stowed position according to an example embodiment of the present disclosure;
(19) FIG. **18** illustrates the view of FIG. **17** with the folding blade in the folded, stowed position according to an example embodiment of the present disclosure;

(20) FIG. **19** illustrates a view of a folding tool including a folding blade and a handle with a piece removed and the folding blade in a partially closed position according to an example embodiment of the present disclosure;
(21) FIG. **20** illustrates the view of the folding tool of FIG. **19** including a folding blade and a handle with a piece removed and the folding blade in a folded, stowed position according to an example embodiment of the present disclosure; and
(22) FIG. **21** illustrates the view of the folding tool of FIG. **20** with the piece of the handle portion replaced according to an example embodiment of the present disclosure.

DETAILED DESCRIPTION

(23) The present inventions now will be described more fully hereinafter with reference to the accompanying drawings, in which some, but not all embodiments of the inventions are shown. Indeed, these inventions may be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will satisfy applicable legal requirements. Like numbers refer to like elements throughout.
(24) Referring now to FIG. **1**, a tool **100** such as a knife or multipurpose tool that includes a handle portion **110** and a folding blade **120** is depicted. While the tool **100** will be described in the context of a folding knife, other types of tools may readily employ components of embodiments of the present disclosure including the inclusion of those components by multipurpose tools and other types of tools that are not considered multipurpose tools. For purposes of illustration, but not of limitation, however, a folding knife employing embodiments of the present disclosure will now be described. Thus, while the tool portion of the illustrated embodiment comprises a folding blade **120**, the tool **100** can include a wide array of types of tools that can be the tool portion, where the tool portion is capable of being folded about an axis from a deployed position to a stowed position relative to a handle portion **110**.
(25) The tool **100** of FIG. **1** includes a handle portion **110**, a folding blade **120**, and a fastener **130** through an axis of rotation of the folding blade relative to the handle portion. FIG. **2** illustrates the tool **100** of FIG. **1** with the folding blade **120** in the stowed, folded position within the handle portion **110**. As shown, the folding blade **120** includes a thumb tab **122** which is received in a recess of the handle portion and enables a user to use the thumb of the hand holding the tool to move the folding blade from the folded, stowed position of FIG. **2** to the open, deployed position of FIG. **1** using only one hand. Folding tools of various embodiments, such as multipurpose tools and folding knives as depicted employ tools, blades, or other components that are movable between a stowed, folded position to a deployed, open position. Components of folding tools including multipurpose tools can include screw drivers, saws, pliers, scissors, can/bottle openers, etc. The folding capability of these components can be employed for a variety of practical purposes. In the case of a multipurpose tool, having the ability to individually unfold and deploy a tool, blade, or other component can enable the use of that deployed component while remaining components can remain in the stowed, folded position within the handle portion. This folding mechanism enables the deployed component to be used without the various other folding components of the multipurpose tool interfering.
(26) In the deployed, unfolded position, the folding blade, such as the folding blade **120** of FIG. **1** relative to the handle portion **110**, can be safely used by a user holding the handle portion. The nature of folding blades and tools renders them convenient for a wide variety of functions and implementations. Generally, the component in the deployed, unfolded position is used by a user to accomplish a task that imparts forces on the deployed component. In the instance of a folding blade **120** depicted in FIG. **1**, the use may generally involve cutting of a material with the sharp edge **125** of the folding blade.
(27) To improve the safety of use of a folding component, particularly a folding blade of a folding tool, embodiments provided herein include a lock to secure the folding blade in the deployed

position. Securing the folding component in the deployed position enables use of the component without concern for unintended folding of the tool, which can compromise the functionality of the component and injure the hand of a user holding the tool. The lock of embodiments described herein is configured to securely lock a component of a tool, such as a blade as illustrated, in a deployed position.

(28) FIG. **3** illustrates the tool **100** of FIGS. **1** and **2** from a different view, where a portion of the locking mechanism is visible within the handle portion **110**. The visible components of the locking mechanism include a button **140** and one embodiment of a spring **145** in the form of a leaf spring. Also shown in FIG. **3** are a first handle piece **112** and second handle piece **114** of the handle portion. The first handle piece **112** and second handle piece **114** are on opposite sides of the handle portion **110** and are the parts of the handle portion gripped by a user when using the tool **100**. The first handle piece **112** and second handle piece **114** define between them a cavity of the tool **100**, wherein one or more tools, blades, or components of a tool or multipurpose tool may be disposed when in the folded, stowed position.

(29) FIG. **4** illustrate the tool **100** of FIGS. **1-3** with the second handle piece **114** of FIG. **3** removed and viewed from a side in which the button **140** is facing. As such, only a portion of the button **140** is visible. The button **140** of the illustrated embodiment may be attached to or formed with the locking wedge **150**, and the locking wedge and button may be secured to the spring **145**. The removal of the second handle piece **114** along with omission of various parts in the following drawings is for visibility of various parts of the tool and better understanding of the operation of the illustrated parts. Also shown in FIG. **4** is the axis **135** about which the folding blade **120** rotates between the folded, stowed position and the depicted deployed position.

(30) The spring **145** is depicted as a leaf spring secured to a chassis of the handle portion **110** and/or to the first handle piece **112** or second handle piece **114**. The spring **145** may be attached to a component of the handle portion **110** with or without the use of fasteners. In the illustrated embodiment, two locating features are illustrated which serves to secure the spring **145** within the handle portion **110** and preclude rotation of the spring within the handle portion **110**. The illustrated locating features include a lug **147** and a stop **148**. The lug **147** provides a locating mechanism for locating the spring **145** within the handle portion **110**, while the stop **148** prevents any relative movement between the spring **145** and the lug **147** or the stop **148**. The spring **145** of the illustrated embodiment is a separate piece from the handle portion **110**, though in some embodiments a leaf spring may be integral to the frame and formed from a piece of the handle portion **110**. The spring of example embodiments can also be a coil spring, such as a coil spring disposed within or attached to the button **140**.

(31) Also visible within FIG. **4** is the locking wedge **150**. The locking wedge **150** is configured to engage at least a portion of a locking surface **165** of a cam portion **160** of the blade **120**. The spring **145** serves to bias the locking wedge **150** toward a locked position where the locking wedge **150** engages a locking surface of the cam portion **160**. The locking surface is substantially parallel to the axis **135**. The button **140** is used to push against the bias of the spring and to move the locking wedge **150** out of contact with the locking surface of the cam portion **160**.

(32) FIG. **5** illustrates another view of the tool **100** of FIG. **4**. As shown from a different angle, the locking wedge **150** and button **140** are integrally formed, and in contact with the spring **145**. The spring biases the locking wedge **150** and button **140** into engagement with the cam portion **160**. The blade **120**, in the deployed position, engages the locking wedge **150** at the locking surface **165** of the cam portion **160**. The locking wedge **150** becomes “wedged” between the locking surface **165** and a stop surface **163** of abutment **170**. Also when the blade **120** is in the open, deployed position, a stopping surface **167** of the blade engages another surface of the abutment **170**. In this manner, the blade is precluded from rotating about axis **135** in a clockwise direction by the stopping surface **167** of the blade engaging the abutment **170**, and precluded from rotating about the axis in a counter-clockwise direction by the locking surface **165** of the cam portion **160**

engaging the locking wedge **150**, and the locking wedge **150** being wedged between the locking surface **165** and the stop surface **163** by the abutment **170**.

(33) According to some embodiments, the locking wedge **150** can be secured to the button **140**, such as by a fastener. In other embodiments the button **140** is integrally formed with the locking wedge **150**. When the button **140** is integrally formed with the locking wedge **150**, the alignment between them is precise. Further, the button **140** and locking wedge **150** integrally formed together form a larger component that is able to float within the handle portion as described further below, where the integrally formed button **140** and locking wedge **150** are not attached to other components.

(34) As shown in FIG. **6**, which is another view of the tool **100** in a section view, with the section line taken through a center of the button **140** with the blade **120** omitted. As depicted, the locking wedge **150** and button **140** can float between the first handle piece **112** and the second handle piece **114**, with portions of one or both handle pieces forming a channel within which the button **140** and locking wedge **150** can move, while the locking wedge **150** is held between the first handle piece **112** and second handle piece **114**. A first aperture **116** in the first handle piece **112** can receive and align the locking wedge **150**, while a second aperture **118** in the second handle piece **114** can receive and align the button **140**. This arrangement both captures the connected locking wedge **150** and button **140** within the handle portion, but also enables the floating of the components while maintaining proper alignment. The spring **145** is illustrated biasing the button toward the second handle piece **114**, where the locking wedge **150** is in position to engage the locking surface **165** of the cam portion **160** of the blade.

(35) FIG. **7** illustrates another view of the tool **100** shown in FIG. **4** with the first handle piece **112** removed. As shown, the spring **145** biases the locking wedge **150** is in a locked position in contact with the locking surface **165** of the cam portion **160** of the folding blade **120**. The locking wedge **150** is engaged in the locked position between the locking surface **165** and abutment **170**. The abutment **170** also engages stopping surface **167** of the cam portion **160** such that the blade in the illustrated embodiment of FIG. **7** is precluded from clockwise rotation by the locking wedge **150** which is also pressed against the abutment and precluded from counterclockwise rotation by the stopping surface **167** engaging the abutment **170**. With the locking wedge **150** in this locked position, the folding blade **120** cannot be rotated about axis **135** from the illustrated deployed position to the folded, stowed position.

(36) FIG. **8** illustrates the locking wedge **150** in the unlocked position with the button pressed to drive the locking wedge **150** against the spring bias of the spring to the unlocked position, with the spring **145** removed for ease of understanding. However, visible in FIG. **8** is the button **140** which is used to press the spring **145** against its biasing force to move the locking wedge from the locked position, where the locking wedge **150** engages the locking surface **165** of the cam portion **160** of the folding blade **120**. The spring **145** biases the lock wedge toward the cam portion **160** of the folding blade **120** such that when the folding blade is in the deployed position, the spring **145** pushes the locking wedge **150** to engage the locking wedge with the locking surface **165**.

(37) FIG. **9** illustrates the components shown in FIG. **8** with the locking wedge **150** engaging the locking surface **165** of the cam portion **160** of the folding blade **120**. As shown, the locking wedge **150** engages the locking surface **165** and precludes rotation of the folding blade **120** about the axis **135** in the clockwise direction (of the view of FIG. **8**) toward the folded, stowed position. This locking mechanism maintains the folding blade **120** in the deployed position for effective and safe use.

(38) FIG. **10** illustrates a top view of the arrangement of components of FIG. **9** with the locking wedge **150** in the locked position relative to the cam portion **160** of the folding blade **120**. As depicted, there is a degree of overlap **152** between the locking wedge **150** and the cam portion **160**. This overlap provides the locking function of the locking mechanism described herein. FIG. **11** illustrates a different view of the locking mechanism with the handle portion removed. As shown,

the locking wedge **150** is engaged in a locked position with the locking surface **165** of the cam portion **160** of the folding blade. The degree of overlap **152** is also illustrated by the broken lines. Also shown is thumb tab **122** of the folding blade **120**, which can be used to move the blade from the folded, stowed position to the deployed position with the use of only one hand in some embodiments.

(39) The locking wedge **150** described herein includes a specifically shaped top surface **154** that enables secure engagement between the top surface **154** of the locking wedge and the locking surface **165** of the cam portion **160** of the folding blade. According to an example embodiment, the locking surface **165** and cam portion **160** can be specified with dimensional tolerances that are well within manufacturing capabilities of readily available machine tools, such that the locking mechanism can be readily produced at a reasonable cost. The configuration of the top surface **154** of the locking wedge **150** enables these cost-effective tolerances on the cam portion **160**.

(40) The components within detail circle **200** of FIG. **11** are illustrated enlarged in FIG. **12**. As shown, the locking wedge **150** is in the locking position relative to locking surface **165** of the cam portion **160** of the folding blade. The top surface **154** overlaps with the locking surface by a length shown as overlap **152**. Arrow **210** illustrates the direction of the biasing force of the spring **145** shown in FIGS. **3-5**. Also shown in FIG. **12** is the aperture **116** through the first handle piece **112** configured to receive and align the locking wedge **150** therein, such that the aperture **116** permits greater lateral travel of the locking wedge and the button **140** than if the locking wedge were confined between the first handle piece **112** and the second handle piece **114**. The aperture **116** can further enable the locking wedge **150** to be “floating” within the handle portion **110**, as the aperture can align the locking wedge and restrict movement in directions other than parallel to the axis of rotation of the folding tool.

(41) According to some embodiments, the button **140** may be integrally attached to the locking wedge **150** such that a biasing force applied to the button **140**, such as with a leaf spring or coil spring, conveys the biasing force to the locking wedge. Alternatively, the spring, such as leaf spring **145**, may be attached to the locking wedge, such as with a fastener, to bias the locking wedge directly.

(42) According to the illustrated embodiment of FIG. **12**, a portion of the top surface **154** of the locking wedge that engages the locking surface **165** is an arcuate surface. The portion of the top surface **154** of the locking wedge **150** that is not arcuate serves, in part, to maintain alignment of the locking wedge **150** within aperture **116**. According to an example embodiment, a cam portion **160** of a folding blade **120** may be around ⅛th inch or 0.125 inches, such that the locking surface may be of around the same width. FIG. **12** also includes detail circle **195**, where FIG. **13** illustrates a detail view of detail circle **195**.

(43) The illustrated example of FIG. **13** is one embodiment of how the locking wedge may be configured; however, the top surface can be configured with an arced surface of various radii and/or a straight, angled surface. According to the illustrated embodiment of FIG. **13**, the locking wedge may include an arcuate surface having a radius of about ⅞th inch or 0.875 inches. In this way, the angle of slope of the top surface **154** is 0.6 degrees at an entry point to the locking wedge **150** top surface **154**, 6.1 degrees at 0.6 inches from the entry point, and 12.6 degrees at 0.125 inches from the entry point. The slope of the arcuate top surface **154** permits greater flexibility in tolerances of the cam portion **160** of the folding blade **120** as explained further below. The arc of the locking wedge **150** top surface **154** more effectively engages the locking surface **165** at a range of engagement depths. Provided the coefficient of friction is greater than a sine of the angle of the top surface **154**, the locking surface **165** should maintain engagement in the locked position. The top surface **154** can be of a higher friction material or texture to aid the locking mechanism.

(44) According to another embodiment, the arced surface may include a straight section, such as at an entrance to the arced surface, where the straight section is, for example, sloped at 3.3 degrees. Such an angle corresponds to an angle whereby frictional engagement between the locking wedge

**150** and the locking surface **165** always exceeds the force component in a direction causing the lock to move relative to the blade. Adding such a straight section can improve clearance for the locking wedge **150** to engage the locking surface **165** without reducing performance of the locking mechanism.

(45) FIG. **14** illustrates the tool **100** of the embodiments described above from a different perspective, with the second piece **114** of the handle portion **110** removed and the first handle piece **112** of the handle portion present. The folding blade **120** is depicted in the deployed position. Visible in the view of FIG. **14** is the locking wedge **150**, spring **145**, and button **140**. Also visible is the abutment **170** configured to engage the stopping surface **167** of the folding blade **120**. The abutment **170** of an example embodiment is integrally formed with the first handle piece **112**. The integral formation of the abutment **170** with the first handle piece **112** provides strength to the mechanism which locks and holds the blade **120** in the deployed position, which improves safety of use of the blade.

(46) FIG. **15** depicts another view of the aforementioned components in a plan view, with the cam portion **160** of the folding blade **120** and the abutment **170** shaded for ease of discerning between parts described herein. Also visible in the view of FIG. **15** is a detent ball **151** or raised element positioned on the locking wedge **150**, the purpose of which is further described below. As shown, the folding blade **120** is in the deployed position locked using the locking wedge **150**. The blade rotates clockwise from the stowed position to reach the deployed position illustrated. When the folding blade **120** is fully deployed, the stopping surface **167** of the folding blade **120** engages a surface of the abutment **170**. This engagement prevents the folding blade **120** from rotating any further in the clockwise direction. When the stopping surface **167** engages the abutment **170**, the bias of spring **145** drives the locking wedge **150** into engagement, along the top surface **154** with the locking surface **165** of the cam portion **160**. The arcuate top surface **154** of the locking wedge **150** ensures a secure engagement with the locking surface **165** while the stopping surface **167** engages the abutment **170**. This provides for locking the folding blade **120** in the deployed position with virtually no play in the blade about the axis **135** of rotation.

(47) Manufacturing tolerances of interfacing components generally correlate strongly with cost of manufacturing. Maintaining very tight tolerances can be costly in terms of both the precision machinery required and potential for a higher defect rate leading to scrap. Embodiments described herein avoid such very tight tolerances as the design is resilient to tolerances that are more readily achieved and can be achieved at generally lower costs that high-precision, tight tolerances. The locking wedge **150** of example embodiments avoids requiring such high-precision, tight on the folding blade **120** and the cam portion **160** thereof. This flexibility is in part based on the arcuate top surface **154**. The arcuate top surface **154** securely engages the locking surface **165** at varying degrees of overlap **152**. A nominal design parameter of the position of the locking surface **165** on the cam portion **160** of the folding blade **120** may be configured to engage the arcuate surface at around half of a width of the cam portion **160**, which may be for example 0.06 inches. An engagement overlap **152** of 0.06 inches engages an arcuate top surface **154** having a radius of 0.625 inches at a position where the surface has an angle of 6.1 degrees. If the position of the locking surface **165** changes within a reasonable tolerance (e.g., based on a formed position of the locking surface **165** or of the stopping surface **167**), the locking surface **165** will still obtain sufficient purchase on the locking wedge **150** top surface **154**.

(48) According to some embodiments, an aperture may be present through the first handle piece **112** of the handle portion **110** to provide access to the locking wedge **150**. In this way, a user may be able to access and press the locking wedge **150** further into engagement with the locking surface **165**. The aperture may include a button that a user may press to drive the locking wedge further into engagement with the locking surface **165** to further improve the degree of overlap **152** and to further pinch the folding blade between the locking wedge **150** and the abutment **170**. The aperture additionally provides clearance for a maximum amount of lock travel within the combined handle

portion thickness as the aperture allows the combined locking wedge **150** and button **140** to travel a greater distance than if they were limited by the first handle piece **112** an second handle piece **114**.
(49) Having described the locking mechanism above, the release of the locking mechanism is described herein. As noted above, pressing button **140** drives the spring against its bias force and pushes the locking wedge **150** out of engagement with the locking surface **165**. This position is illustrated, for example, in FIG. **7**. Once the locking wedge **150** has been driven to the unlocked position, the folding blade **120** may be rotated about axis **135** toward the closed position as shown in FIG. **16**. As shown, the locking surface **165** is able to clear the locking wedge **150**. Once the locking surface has at least partially cleared the locking wedge **150**, the button **140** no longer needs to be pressed to continue the closing motion, as the bias of the spring **145** presses the locking wedge against the cam portion **160** of the folding blade, where the locking wedge can ride along the cam portion during the closing motion.
(50) FIG. **17** illustrates the folding blade **120** rotated further about axis **135** toward the folded, stowed position which is depicted in FIG. **18**. The folding blade **120** of the illustrated embodiment includes a recess **141** to accommodate the button **140** and a stop surface **163** which engages the abutment **170** to prevent over rotation of the folding blade **120** past the closed position which may compromise the sharpness of the blade if it were allowed to contact a bottom of a cavity in which it is received in the folded, stowed position.
(51) FIG. **19** illustrates the tool **100** in the same position as found in FIG. **17** with the folding blade **120** partially closed from the opposite side of the tool with the first handle piece **112** of the handle portion **110** removed. The locking wedge **150** of an example embodiment includes a retaining feature, where the retaining feature engages the folding blade **120** to hold and secure the folding blade in the folded, stowed position. According to the illustrated embodiment, the folding blade includes a detent **123**, while the locking wedge **150** includes a retaining feature in the form of a detent ball **151** as illustrated in FIG. **15**. Upon rotating the folding blade **120** to the folded, stowed position as shown in FIG. **20**, the detent ball **151** of the locking wedge **150** is received into the detent **123** to hold the folding blade in the stowed position.
(52) FIG. **21** illustrates the tool **100** with the handle portion **110** intact and the folding blade **120** in the stowed, folded position within a cavity within the handle portion. Also shown is aperture **116** through the first handle piece **112** of the handle portion **110** as described above. The aperture **116** provides access, either directly or indirectly, to the locking wedge **150** such that a user can press the locking wedge further into engagement with a locking surface of the folding blade **120** as appropriate and permits greater travel of the locking wedge **150** and button **140**.
(53) While the illustrated embodiments of FIGS. **1-18** are primarily directed to a folding knife tool, the folding and locking mechanism described herein can be employed for a wide array of folding tools since the mechanism functions regardless of what type of tool on which embodiments are employed including, for example, non-folding tools.
(54) Embodiments provided herein include a tool including: a handle portion; a folding tool portion, wherein the folding tool portion is configured to be rotatable about a folding tool axis between a folded, stowed position at least partially within the handle portion, and a deployed position substantially outside of the handle portion, where the folding tool portion comprises a cam portion disposed about the folding tool axis, where the cam portion defines a locking surface; and a locking element, wherein the locking element is configured to engage the locking surface of the folding tool portion in response to the folding tool portion moving to the deployed position, where the locking element is spring biased by a spring in a direction parallel to the folding tool axis, and where the locking element defines at least one of an arcuate surface or a wedge surface for engaging the locking surface of the folding tool portion.
(55) The handle portion of some embodiments defines a first handle piece and a second handle piece, where the first handle piece defines a first aperture, where the locking element is aligned with and guided by the first aperture. The tool of some embodiments further includes a button,

where the button is at least one of attached to the locking element or integrally formed with the locking element, where the button is configured to drive the locking element against spring bias to disengage the locking element from the locking surface. The button of some embodiments defines a central button axis extending therethrough parallel to the folding tool axis, where the locking element defines a central locking element axis parallel to the folding tool axis, and where the central button axis and the central locking element axis are offset from one another.

(56) According to some embodiments the second handle piece defines a second aperture, where the button is aligned with and guided by the second aperture. The button and the locking element of some embodiments are not mechanically coupled to the handle portion or the spring. According to some embodiments the locking element defines a detent ball, where the cam portion defines a recess, where in the folded, stowed position the detent ball engages the recess and holds the folding tool portion in the folded, stowed position. The handle portion of some embodiments includes an abutment, and the folding tool portion comprises a stopping surface defined by the cam portion, where in the deployed position, the folding tool portion is secured rotationally about the folding tool axis in a first direction by the locking element engaging the locking surface and in a second direction by the stopping surface engaging the abutment. According to some embodiments in the deployed position the locking element is held between the locking surface of the folding tool portion and a stop surface of the abutment. According to some embodiments the handle portion includes a first handle piece and a second handle piece, wherein the abutment is integrally formed with the first handle piece.

(57) Embodiments provided herein include a tool including: a handle portion; a folding tool portion, where the folding tool portion is configured to be rotatable about a folding tool axis between a folded, stowed position at least partially within the handle portion, and a deployed position substantially outside of the handle portion, where the folding tool portion includes a cam portion disposed about the folding tool axis, where the cam portion defines a locking surface and a stopping surface where in the deployed position the stopping surface of the cam portion engages an abutment integrally formed within the handle portion; and a locking element, where the locking element is configured to engage the locking surface of the cam portion in response to the folding tool portion moving to the deployed position, where the locking element is spring biased by a spring in a direction parallel to the folding tool axis.

(58) According to some embodiments the locking element defines at least one of an arcuate surface or a wedge surface for engaging the locking surface of the folding tool portion. The handle portion of some embodiments defines a first aperture therein, wherein the locking element is aligned with and guided by the first aperture. The tool of an example embodiment further includes a button, where the button is at least one of attached to the locking element or integrally formed with the locking element, where the button is configured to drive the locking element against spring bias to disengage the locking element from the locking surface.

(59) According to some embodiments the button defines a central button axis extending therethrough parallel to the folding tool axis, where the locking element defines a central locking element axis parallel to the folding tool axis, and where the central button axis and the central locking element axis are offset from one another. The handle portion of an example embodiment defines a first handle piece and a second handle piece, where the first handle piece defines a first aperture, where the second handle piece defines a second aperture, where the locking element is aligned with and guided by the first aperture, and where the button is aligned with and guided by the second aperture. The locking element of some embodiments defines a detent ball, wherein the cam portion defines a recess, wherein in the folded, stowed position the detent ball engages the recess and holds the folding tool portion in the folded, stowed position.

(60) Embodiments described herein include a tool including: a handle portion; a folding tool portion, where the folding tool portion is configured to be rotatable about a folding tool axis between a folded, stowed position at least partially within the handle portion, and a deployed

position substantially outside of the handle portion, where the folding tool portion includes a cam portion disposed about the folding tool axis, where the cam portion defines a locking surface; and a locking element, where the locking element is configured to engage the locking surface of the cam portion in response to the folding tool portion moving to the deployed position; and a button, where the button is at least one of attached to the locking element or integrally formed with the locking element, where the button is configured to drive the locking element against spring bias to disengage the locking element from the locking surface

(61) According to some embodiments the locking element and button are spring biased by a spring in a direction parallel to the folding tool axis. The button of some embodiments defines a central button axis extending therethrough parallel to the folding tool axis, where the locking element defines a central locking element axis parallel to the folding tool axis, and where the central button axis and the central locking element axis are offset from one another. According to some embodiments the handle portion defines a first handle piece and a second handle piece, where the first handle piece defines a first aperture, where the second handle piece defines a second aperture, where the locking element is aligned with and guided by the first aperture, and where the button is aligned with and guided by the second aperture.

(62) According to certain embodiments the locking element defines a detent ball, where the cam portion defines a recess, where in the folded, stowed position the detent ball engages the recess and holds the folding tool portion in the folded, stowed position. The locking element of some embodiments defines at least one of an arcuate surface or a wedge surface for engaging the locking surface of the folding tool portion, where the at least one of the arcuate surface or the wedge surface defines a texture having a higher coefficient of friction than a smooth surface.

(63) Many modifications and other embodiments of the inventions set forth herein will come to mind to one skilled in the art to which these inventions pertain having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. Therefore, it is to be understood that the inventions are not to be limited to the specific embodiments disclosed and that modifications and other embodiments are intended to be included within the scope of the appended claims. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

## Claims

1. A tool comprising: a handle portion comprising an abutment defining a first stop surface and a second stop surface; a folding tool portion, wherein the folding tool portion is configured to be rotatable about a folding tool axis in a first direction from a folded, stowed position at least partially within the handle portion to a deployed position substantially outside of the handle portion, wherein the folding tool portion comprises a cam portion disposed about the folding tool axis, wherein the cam portion defines a locking surface; and a locking element, wherein the locking element is configured to engage the locking surface of the folding tool portion and become wedged between the first stop surface of the abutment and the locking surface of the folding tool portion in response to the folding tool portion moving to the deployed position, wherein in the deployed position, the folding tool portion is precluded from rotating in the first direction about the folding tool axis by engaging in the second stop surface of the abutment, and precluded from rotating in a second direction about the folding tool axis, opposite the first direction, by the locking element engaging the locking surface of the folding tool portion, wherein the locking element is spring biased by a spring in a direction parallel to the folding tool axis, and wherein the locking element defines at least one of an arcuate surface or a wedge surface for engaging the locking surface of the folding tool portion.

2. The tool of claim 1, further comprising a button, wherein the button is at least one of attached to the locking element or integrally formed with the locking element, wherein the button is configured

to drive the locking element against spring bias to disengage the locking element from the locking surface.
3. The tool of claim 2, wherein the button defines a central button axis extending therethrough parallel to the folding tool axis, wherein the locking element defines a central locking element axis parallel to the folding tool axis, and wherein the central button axis and the central locking element axis are offset from one another.
4. The tool of claim 2, wherein the handle portion defines a first handle piece and a second handle piece, wherein the first handle piece defines a first aperture, wherein the second handle piece defines a second aperture, wherein the locking element is aligned with and guided by the first aperture, and wherein the button is aligned with and guided by the second aperture.
5. The tool of claim 4, wherein the button and the locking element are not mechanically coupled to the handle portion or the spring.
6. The tool of claim 2, wherein the folding tool portion comprises a stopping surface defined by the cam portion, wherein in the deployed position, the folding tool portion is secured rotationally about the folding tool axis in a first direction by the locking element engaging the locking surface and in a second direction by the stopping surface engaging the abutment.
7. The tool of claim 6, wherein the handle portion comprises a first handle piece and a second handle piece, wherein the abutment is integrally formed with the first handle piece.
8. The tool of claim 1, wherein the locking element defines a detent ball, wherein the cam portion defines a recess, wherein in the folded, stowed position the detent ball engages the recess and holds the folding tool portion in the folded, stowed position.
9. The tool of claim 1, wherein the locking element defines the arcuate surface, wherein the arcuate surface of the locking element engages the locking surface of the folding tool portion at different angles based on a degree of overlap between the locking surface and the arcuate surface.
10. A tool comprising: a handle portion comprising a first handle piece and a second handle piece, the first handle piece defining an abutment; a folding tool portion, wherein the folding tool portion is configured to be rotatable about a folding tool axis in a first direction from a folded, stowed position at least partially between the first handle piece and the second handle piece to a deployed position substantially outside of the handle portion, wherein the folding tool portion comprises a cam portion disposed about the folding tool axis, wherein the cam portion defines a locking surface; a locking element, wherein the locking element is configured to be captured between the locking surface of the folding tool portion and the abutment in response to the folding tool portion moving to the deployed position, wherein in the deployed position further rotation in the first direction about the folding tool axis is precluded by the folding tool portion engaging the abutment, wherein the locking element is spring biased by a spring in a direction parallel to the folding tool axis, wherein the locking element defines at least one of an arcuate surface or a wedge surface for engaging the locking surface of the folding tool portion; and a button defining a central button axis extending therethrough parallel to the folding tool axis, wherein the locking element defines a central locking element axis parallel to the folding tool axis, wherein the central button axis and the central locking element axis are offset from one another, wherein the button is at least one of attached to the locking element or integrally formed with the locking element, and wherein the button is configured to drive the locking element against spring bias to disengage the locking element from the locking surface.
11. The tool of claim 10, wherein the handle portion defines a first handle piece and a second handle piece, wherein the first handle piece defines a first aperture, wherein the second handle piece defines a second aperture, wherein the locking element is aligned with and guided by the first aperture, and wherein the button is aligned with and guided by the second aperture.
12. The tool of claim 11, wherein the button and the locking element are not mechanically coupled to the handle portion or the spring.
13. The tool of claim 10, wherein the locking element defines a detent ball, wherein the cam

portion defines a recess, wherein in the folded, stowed position the detent ball engages the recess and holds the folding tool portion in the folded, stowed position.

14. A tool comprising: a handle portion comprising a first handle piece defining a first aperture and a second handle piece defining a second aperture; a folding tool portion, wherein the folding tool portion is configured to be rotatable about a folding tool axis between a folded, stowed position at least partially between the first handle piece and the second handle piece, and a deployed position substantially outside of the handle portion, wherein the folding tool portion comprises a cam portion disposed about the folding tool axis, wherein the cam portion defines a locking surface; a locking element, wherein the locking element is received within the first aperture wherein the locking element is movable along a locking axis parallel to the folding tool axis between an engaged position in which the locking element is engaged with the locking surface of the folding tool portion in the deployed position, and a disengaged position where the locking element does not contact the locking surface; and a button, wherein the button is received within the second aperture and is accessible through the second handle piece, wherein the button is at least one of attached to or integrally formed with the locking element, and wherein the button is movable along an axis parallel to the locking axis against a spring bias to move the locking element to the disengaged position, wherein the button defines a central button axis extending therethrough parallel to the folding tool axis, wherein the locking element defines a central locking element axis parallel to the folding tool axis, and wherein the central button axis and the central locking element axis are offset from one another.

15. The tool of claim 14, wherein the locking element defines a detent ball, wherein the cam portion defines a recess, wherein in the folded, stowed position the detent ball engages the recess and holds the folding tool portion in the folded, stowed position.

16. The tool of claim 14, wherein a surface of the locking element that engages the locking surface comprises a surface texture configured to engage the locking surface with a higher friction than a smooth surface texture.